

Pell Grant Share and Graduate Earnings in US Four-Year Institutions

Statistical Analysis Report

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Overview

This report analyses whether the share of undergraduates who receive Pell Grants is associated with the median earnings of former students ten years after they first enrolled, among US institutions that mainly award bachelor's degrees. I used the College Scorecard, a dataset published by the U.S. Department of Education and listed on Data.gov (<https://catalog.data.gov/dataset/college-scorecard-c25e9>; data portal: <https://collegescorecard.ed.gov/data>). The question matters because earnings are widely used to judge the value of institutions, so it is important to understand how they relate to the students an institution serves. The analysis combines descriptive statistics, four visual models and a Spearman rank correlation test.

Dataset Description

The College Scorecard provides one row per US higher education institution, with measures of admissions, costs, completion, student debt and earnings (U.S. Department of Education, n.d.). The most recent institution-level file contains 6,273 rows and 3,308 columns, from which I extracted 23 columns relevant to the question. After restricting the data to 1,947 institutions whose predominant award is the bachelor's degree, 1,787 had values for both of the key variables. These are the share of undergraduates receiving Pell Grants (a proportion between 0 and 1) and median earnings ten years after entry (in US dollars). I also used the institution's control type (public, private nonprofit or private for-profit), the completion rate, the cost of attendance, the admission rate and undergraduate enrollment.

Methods

Data preparation and descriptive statistics. Before testing anything I screened the data, an initial data analysis step that checks data quality and documents every decision before the main analysis begins (Lusa et al., 2024). The screening found text entries reading PrivacySuppressed in the debt column, which I treated as missing values rather than numbers, and blank earnings values for some institutions, which I did not impute because the missingness is probably not random. It also found that campuses of one university system or chain share an identical earnings value, which affects the independence assumption discussed below. I restricted the sample to bachelor's-granting institutions so that the institutions compared are similar in type. Because earnings and Pell share are skewed, I summarised them with medians and interquartile ranges as well as means, and I counted institutions by control type to show the balance of the sample.

Visual models. I built four charts, each chosen to answer a different question, using Matplotlib and Seaborn (Waskom, 2021). A histogram shows the distribution of earnings and whether a test that assumes normality would be suitable. A boxplot compares earnings across control types and reveals

outliers. A scatterplot with a smoothed curve shows the direction and shape of the relationship between Pell share and earnings, which is the question being asked. A heatmap of rank correlations shows how Pell share and earnings relate to other institutional variables.

Hypothesis test. The null hypothesis is that there is no monotonic association between Pell share and median earnings (the population Spearman correlation is zero) and the alternative is that there is one. I used a two-sided Spearman rank correlation test at the 0.05 level. This test suits two continuous variables whose relationship is consistently increasing or decreasing but not necessarily linear, and it is preferred over the Pearson coefficient when data are not normally distributed (Schober et al., 2018). The histogram showed that earnings are right-skewed and the scatterplot showed a decreasing, curved pattern, so the rank-based test was the better choice. Choosing the test after screening the data is consistent with the view that analysis plans may be adjusted in light of what initial data analysis reveals, provided that the change is documented and justified (Lusa et al., 2024). The test assumes that observations are paired, that both variables are at least ordinal, that the relationship is monotonic and that observations are independent. The first three are supported by the data and the figures. Independence is doubtful because campuses of the same system share one earnings value, so I repeated the test keeping one institution for each repeated value. The test was run with SciPy (Virtanen et al., 2020). A p-value indicates how incompatible the data are with the null hypothesis but does not measure the size or importance of an effect (Wasserstein & Lazar, 2016), so I also report the correlation coefficient and a bootstrap confidence interval.

Results

Median earnings across the 1,787 institutions have a median of 53,800 US dollars, with half of the institutions between 45,389 and 63,435 dollars, and the distribution is right-skewed (Figure 1). Public and private nonprofit institutions have almost identical median earnings (about 54,700 and 54,100 dollars), while private for-profit institutions have a much lower median of 40,092 dollars (Figure 2). Institutions with a higher share of Pell Grant recipients have lower earnings: the median earnings of institutions with a Pell share up to 0.2 is about 70,000 dollars, falling to about 30,500 dollars where the share exceeds 0.8, and the fall is steepest at low Pell shares (Figure 3). Pell share is also negatively related to the completion rate (rank correlation of -0.62) and to the cost of attendance (-0.46), and completion rate and earnings are positively related (0.59), so these variables overlap (Figure 4).

The Spearman test gave a correlation of -0.550 with a p-value far below 0.001 ($n = 1,787$), and a 95 percent bootstrap interval from -0.586 to -0.510. The null hypothesis of no association is therefore rejected. By the commonly used verbal guide, a coefficient of this size is moderate (Schober et al., 2018). The association is negative within each control type, with correlations of -0.41 for public, -0.61 for private nonprofit and -0.28 for private for-profit institutions. When one institution was kept for each repeated earnings value (1,612 institutions), the correlation was -0.561, almost the same as the full result.

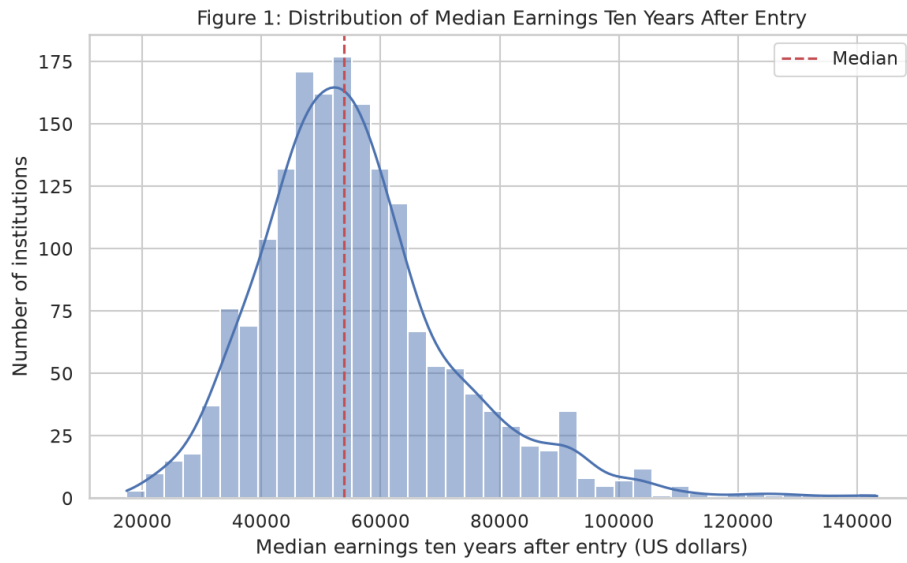


Figure 1. Distribution of median earnings ten years after entry.

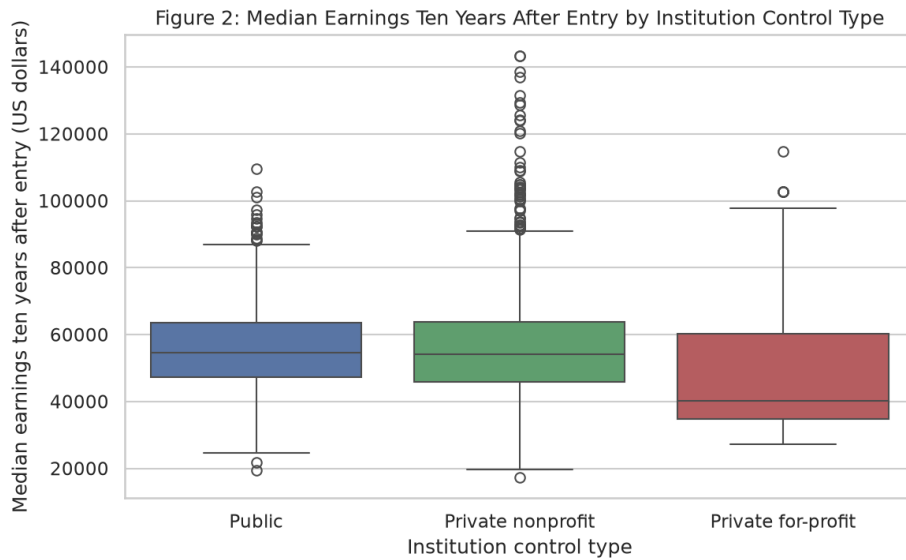


Figure 2. Median earnings by institution control type.

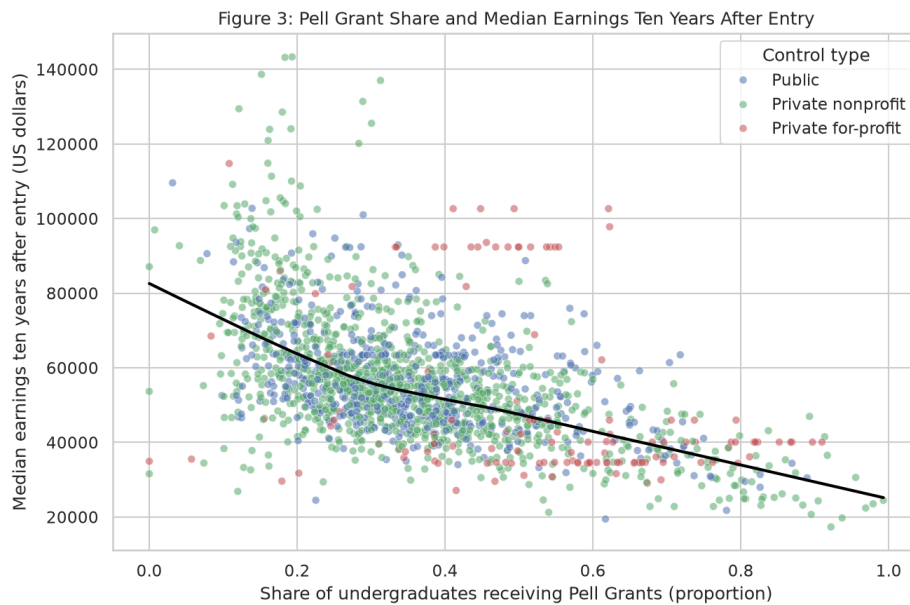


Figure 3. Pell Grant share and median earnings ten years after entry.

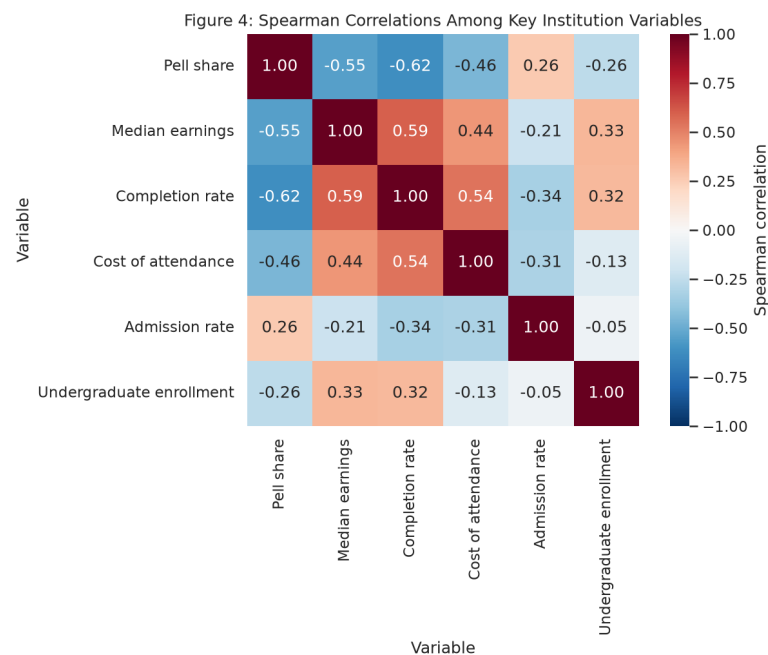


Figure 4. Spearman correlations among key institution variables.

Interpretation for a Non-Technical Audience

Pell Grants are federal grants for students who have financial need, so the share of students receiving them is a rough sign of how many students at a college come from lower-income families. The analysis found that colleges where more students receive Pell Grants tend to have lower earnings among former students ten years after they started. The link is moderate, not perfect. Many colleges with high Pell shares have graduates who earn well, and many with low Pell shares do not, so knowing a college's Pell share only gives a rough guide to its earnings. The finding describes colleges as a whole, not individual people, and it does not show that receiving a Pell Grant causes lower earnings. Other things differ between these colleges, such as the subjects they teach, where they are located and how selective they are, and these could explain part of the pattern. The result should be read as a pattern worth

questioning, and not as a verdict on any college or student.

Limitations and Potential Bias

Dataset limitations. The earnings figures describe only former students who received federal financial aid and were working and not enrolled ten years after entry (National Association of Independent Colleges and Universities, n.d.), so they do not describe all students at an institution. Earnings are also missing for about eight percent of the bachelor's-granting institutions, and I could not tell from the file why. The data are observational, so the analysis can show association but not cause. In addition, the data are reported for institutions, and correlations calculated on groups can be larger than, and cannot substitute for, the correlations between the same properties measured on individuals (Robinson, 1950). Finally, campuses of one system share a single earnings value, so the 1,787 observations contain less independent information than the count suggests.

Potential bias. Missing and suppressed values are a source of bias. Data that do not meet reporting standards are suppressed in the Scorecard, and institutions with small student groups are probably more likely to be missing, so the institutions analysed may not represent all institutions. There is also a risk of bias in interpretation: Pell share, completion, cost and earnings are related to each other, so attributing the pattern to Pell share alone would be misleading. A further ethical concern is misuse. If earnings are used to rank or fund institutions without accounting for the students they serve, institutions that enrol many lower-income students could be unfairly judged. I limited the sample to bachelor's-granting institutions, which also means the conclusions do not apply to community colleges or certificate programmes.

References

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